

# Yuna Park

Irvine, CA | botw461@gmail.com | +1(949)-331-8640 | github.com/Yuna616 | linkedin.com/in/yunapark616

## SUMMARY

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Software Engineer with hands-on experience deploying AI models on Qualcomm Hexagon NPU hardware and building real-time firmware (FreeRTOS/ESP32).

## EXPERIENCE

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### KMU GLOBAL PBL

Irvine, CA

#### Software Development Track

Mar 2026 - Aug 2026

- Selected as a top 1% fellow at Kookmin University for a US-based program building cross-functional products across Applied AI, serverless infrastructure, and full-stack cloud deployments.

### TEACHING ASSISTANT

Seoul, South Korea

#### Paid TA for a Computer Networks Course

Mar 2025- Jul 2025

- Mentored 30+ undergraduates in a Computer Networks course by simplifying complex technical concepts (TCP/IP, routing, network security) during weekly office hours to support academic performance.

## PROJECTS

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### ON-DEVICE AI CHEATING DETECTION SYSTEM

Irvine, CA

#### Embedded Systems Engineer

Jun 2026 - Present

- Architected a **dual-NPU edge AI system** on Radxa Airbox (Qualcomm QCS9075), isolating vision inference (NPU0) and LLM judgment (NPU1) for cloud-independent, real-time execution.
- Deployed **4 quantized vision models** with **full NPU acceleration** (ONNX Runtime QNN EP) and **ported ByteTrack to dependency-free Rust** for optimized object tracking.
- Engineered a **gaze calibration pipeline** (MAD outlier rejection, EMA stabilization) and benchmarked 5 estimation models, achieving 7.10° MAE and a 66% reduction in false positives.

### CARPYBARA - in-vehicle IoT device

Irvine, CA

#### Firmware Engineer

Mar 2026 - Jun 2026

- Architected **FreeRTOS firmware on ESP32-S3**, optimizing system responsiveness by isolating sensor/network tasks (Core 0) from 60 FPS display rendering (Core 1) using mutex-protected shared state.
- Developed an end-to-end sensor fusion pipeline, integrating GPS/IMU telemetry via I2C/UART with complementary filters and transmitting data to a React/Supabase dashboard via BLE/GATT.

### POINT - AI presentation coaching platform

Irvine, CA

#### Full-Stack Developer

Mar 2026 - Jun 2026

- Architected a full-stack AI platform (React/TS, Supabase, Vercel) integrating a 6-agent pipeline (GPT-4o, Whisper, MediaPipe) with a 3-tier feedback queue.

## EDUCATION

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### KOOKMIN UNIVERSITY

Seoul, Korea

Bachelor of Software Engineering - GPA: 4.19/4.5

Mar 2023 - Aug 2027 (Expected)

**Relevant Courses:** Operating Systems, Computer Architecture, System Software, Cloud Computing, Database, C++ Programming, Data Science, Computer Network, File Processing, Algorithms, Object-Oriented Analysis & Design

## SKILLS

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- Embedded & IoT:** ESP32, FreeRTOS, Arduino, Linux, I2C/UART/BLE, PlatformIO
- Edge & Applied AI:** Qualcomm NPU, ONNX Runtime, QAI RT Genie, Computer Vision, LLM/RAG, Multi-Agent
- Full-Stack & Cloud:** TypeScript/JavaScript, React, FastAPI, Supabase, AWS, Vercel
- Languages & Tools:** C, C++, Python, Rust, Java, Git